DATA WAREHOUSING AND MINING LABORATORY

Experiment No. 8: Apply K-Means Clustering on a Dataset and Interpret the Generated Clusters

Aim: To apply the K-Means clustering algorithm using WEKA and interpret the clusters generated from a student performance dataset.

Objectives: After successful completion of this experiment, students will be able to:

1. Understand the concept of clustering.
2. Understand unsupervised learning.
3. Apply the K-Means clustering algorithm using WEKA.
4. Generate clusters from unlabeled data.
5. Interpret cluster centroids.
6. Analyze cluster distributions.
7. Understand the importance of clustering in data mining applications.

Importance of this Experiment:

Unlike classification techniques, clustering does not require predefined class labels. Clustering automatically groups similar records based on their attribute values. Applications of clustering include:

- Customer Segmentation
- Educational Data Mining
- Recommendation Systems
- Healthcare Analytics
- Market Research
- Fraud Detection

This experiment introduces students to unsupervised learning and illustrates how hidden patterns can be discovered from data.

Expected Outcome: After completing this experiment, students will be able to:

1. Apply K-Means clustering using WEKA.
2. Generate clusters from unlabeled datasets.
3. Interpret cluster centroids.

4. Analyze cluster sizes.
5. Understand the concept of unsupervised learning.
6. Identify natural groupings within datasets.

Facilities Required:

Hardware Requirements:

- Personal Computer / Laptop
- Intel i3 Processor or higher
- Minimum 4 GB RAM
- Internet Connection (Optional)

Software Requirements:

- Windows 10/11
- WEKA Tool
- Student Clustering Dataset (student_clustering.arff)

Prerequisite Knowledge: Students should be familiar with:

- Experiment No. 1: Installation and Exploration of WEKA Tool and Available Datasets
- Experiment No. 2: Create and Analyze ARFF Files Using WEKA
- Experiment No. 3: Data Cleaning and Missing Value Treatment
- Experiment No. 4: Data Transformation and Normalization
- Experiment No. 5: Generate Association Rules Using Apriori Algorithm
- Experiment No. 6: Generate Frequent Patterns Using FP-Growth Algorithm and Compare with Apriori
- Experiment No. 7: Apply Naive Bayes and Decision Tree Classification on a Dataset
- Basic Data Mining Concepts

Theory:

1. Clustering

Clustering is a data mining technique used to group similar records into clusters. Records belonging to the same cluster are more similar to each other than to records in other clusters. Clustering helps in discovering hidden structures present in data.

2. Unsupervised Learning

Clustering belongs to the category of Unsupervised Learning. In unsupervised learning:

- No predefined class labels exist.
- The algorithm identifies patterns automatically.
- Similar records are grouped based on attribute values.

Unlike classification, there is no target attribute.

3. K-Means Clustering

K-Means is one of the most widely used clustering algorithms. The algorithm partitions data into K clusters, where, K = Number of desired clusters. Each cluster is represented by its centroid.

4. Centroid

A centroid is the center of a cluster. It represents the average values of all records belonging to that cluster.

Example:

Cluster Centroid

Attendance = 89

Marks = 84

This centroid represents the characteristics of the students belonging to that cluster.

5. Working of K-Means Algorithm

K-Means operates as follows:

Select K Centroids → Assign Records to Nearest Centroid → Calculate New Centroids → Reassign Records → Repeat until Stable

The process continues until cluster assignments no longer change.

6. Advantages of K-Means

- Simple to understand
- Easy to implement
- Fast execution
- Suitable for large datasets
- Effective for exploratory analysis

7. Classification vs Clustering

Feature	Classification	Clustering
Learning Type	Supervised	Unsupervised
Class Labels	Required	Not Required
Target Attribute	Present	Absent
Example Algorithm	Naive Bayes, J48	K-Means
Output	Predicted Class	Cluster Groups

Industry Insight: Clustering is extensively used in customer segmentation, recommendation systems, healthcare analytics, educational analytics, and marketing applications to discover naturally occurring groups within data.

Mini Real-World Example:

Customer Segmentation

Consider the following customers:

Customer	Income	Spending
C1	High	High
C2	Medium	Medium
C3	Low	Low

K-Means may group them into:

1. Premium Customers
2. Regular Customers
3. Budget Customers

without requiring predefined labels.

Food Recommendation System

Children may be grouped using attributes such as:

- Age
- Weight
- Height
- Nutritional Indicators

Possible clusters:

1. Normal Nutrition Group

2. Moderate Risk Group
3. High Nutritional Risk Group

These clusters can support targeted food recommendation strategies.

Dataset Used in This Experiment:

Filename: student_clustering.arff

Dataset Contents:

```
@relation student
@attribute attendance numeric
@attribute marks numeric
@data
90,85
88,83
92,87
70,65
68,60
72,62
40,35
45,30
38,40
```

Dataset Information:

- Relation Name: student
- Instances: 9
- Attributes: 2
- Class Attribute: Not Available
- Learning Type: Unsupervised

Results Obtained During Experiment:

Clustering Algorithm: SimpleKMeans

Number of Clusters: 3

Number of Iterations: 2

Final Cluster Centroids

Attribute	Cluster 0	Cluster 1	Cluster 2
Attendance	89	55.5	92
Marks	84	48.67	87

Cluster Distribution

Cluster	Instances	Percentage
Cluster 0	2	22%
Cluster 1	6	67%
Cluster 2	1	11%

Interpretation of Clusters

Cluster 0

Centroid:

Attendance = 89

Marks = 84

Interpretation: High-Performing Students

Cluster 1

Centroid:

Attendance = 55.5

Marks = 48.67

Interpretation: Average and Low-Performing Students

Cluster 2

Centroid:

Attendance = 92

Marks = 87

Interpretation: Outstanding Student

Important Observation

The generated clusters may not exactly match human expectations because K-Means groups records solely based on similarity and distance calculations. Different initial centroids may produce different cluster assignments. This demonstrates an important characteristic of unsupervised learning.

Understanding the Output:

Clustered Instances

Cluster 0 = 2 Instances

Cluster 1 = 6 Instances

Cluster 2 = 1 Instance

This indicates how many records were assigned to each cluster.

Cluster Centroids

Centroids represent the average profile of records belonging to a cluster.

Example:

Attendance = 89

Marks = 84

indicates a group with higher academic performance compared to other clusters.

Procedure:

Activity 1: Load Dataset

Steps:

1. Open WEKA
2. Select Explorer
3. Open the Preprocess tab
4. Click Open File
5. Select student_clustering.arff
6. Click Open

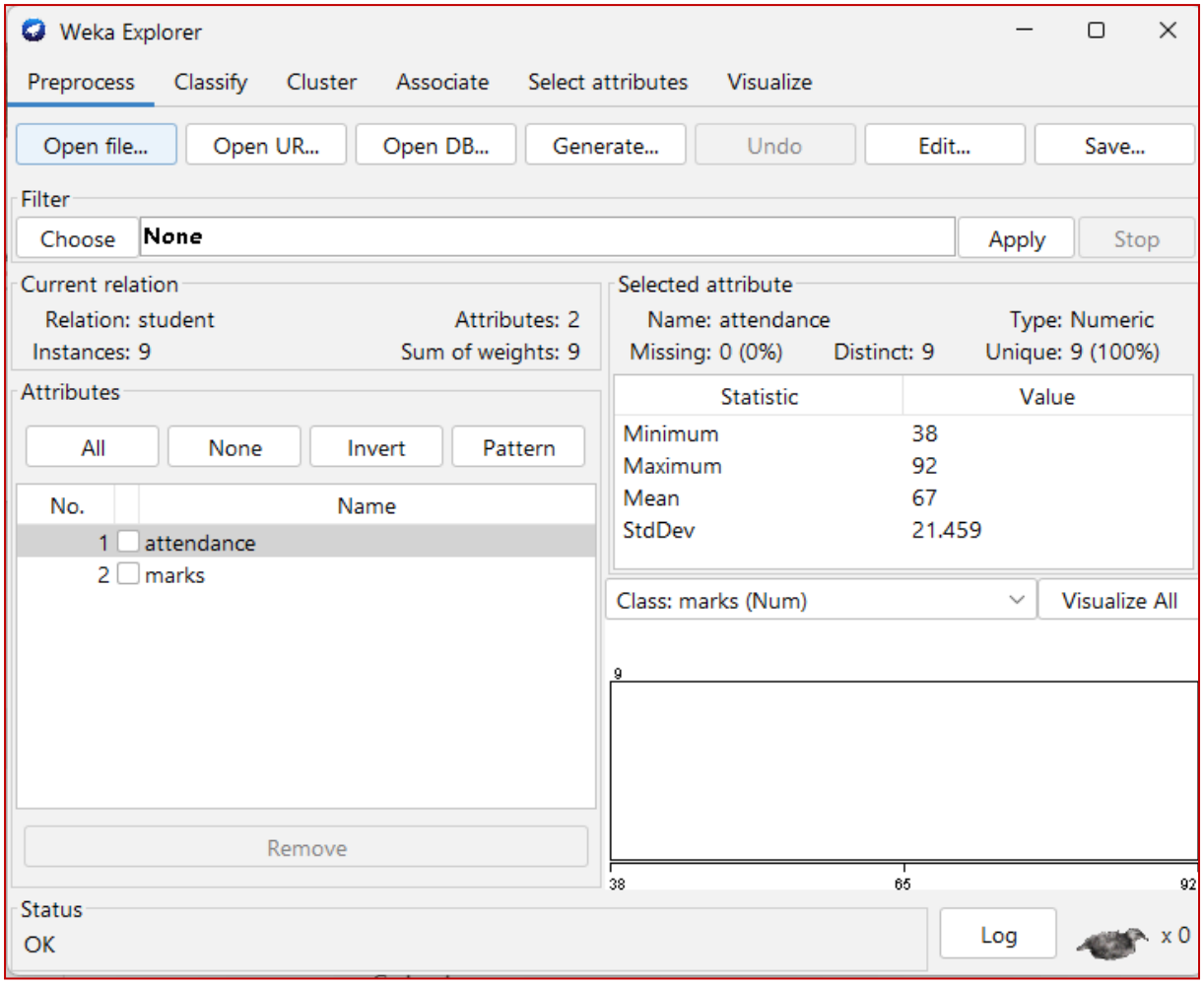


Figure 8.1: Student Clustering Dataset Loaded in WEKA

Activity 2: Open Cluster Tab

Step: Select Cluster tab.

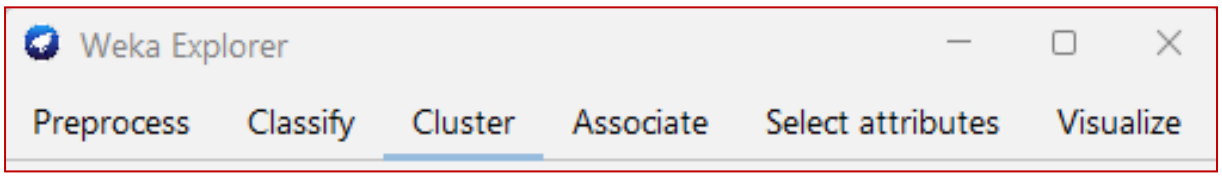


Figure 8.2: WEKA Cluster Interface

Activity 3: Select K-Means Algorithm

Steps:

1. Click: Choose
2. Select: SimpleKMeans
3. Set: Number of Clusters (K) = 3
4. Click OK.

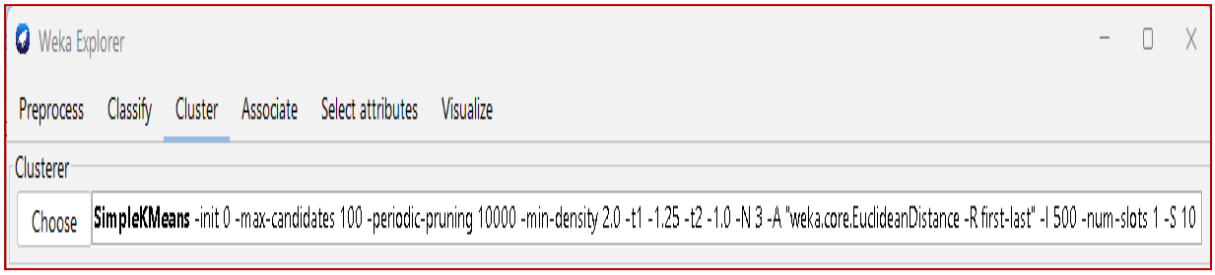


Figure 8.3: Selecting SimpleKMeans Algorithm

Activity 4: Generate Clusters

Steps:

1. Click: Start
2. Observe:
   - Cluster Centroids
   - Cluster Sizes
   - Number of Iterations

```
=== Run information ===

Scheme:     weka.clusterers.SimpleKMeans -init 0 -max-candidates 100 -
periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 3 -A
"weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10
Relation:   student
Instances:  9
Attributes: 2
            attendance
            marks
Test mode:  evaluate on training data


=== Clustering model (full training set) ===


kMeans
======

Number of iterations: 2
Within cluster sum of squared errors: 0.8097890519020098

Initial starting points (random):

Cluster 0: 90,85
Cluster 1: 70,65
Cluster 2: 92,87

Missing values globally replaced with mean/mode

Final cluster centroids:
                        Cluster#
Attribute   Full Data       0       1       2
            (9.0)     (2.0)     (6.0)     (1.0)
==========================================================
attendance       67       89     55.5       92
marks       60.7778       84  48.6667       87



Time taken to build model (full training data) : 0.01 seconds

=== Model and evaluation on training set ===

Clustered Instances

0     2 ( 22%)
1     6 ( 67%)
2     1 ( 11%)
```

Figure 8.4: K-Means Clustering Results

Activity 5: Interpret Clusters

Analyze:

- Cluster Centroids
- Cluster Sizes
- Student Groupings

Record observations

Common Mistakes While Performing Clustering

1. Assuming clustering requires class labels.
2. Confusing clustering with classification.
3. Interpreting clusters without examining centroids.
4. Assuming cluster numbers represent quality rankings.
5. Ignoring the effect of initial centroid selection.

Common Discussion

1. What is clustering?
2. What is unsupervised learning?
3. What is a centroid?
4. How does K-Means work?
5. Why are class labels not required?
6. How are clusters interpreted?
7. Can different runs produce different clusters?
8. What is the significance of centroid values?
9. How is clustering different from classification?
10. Give real-world applications of clustering.

Student Activity:

Create a dataset containing:

- At least 10 records
- At least 2 numerical attributes

Apply: SimpleKMeans using: K = 3

Interpret:

- Cluster Sizes
- Cluster Centroids
- Cluster Meaning

Repeat clustering using: K = 2 and K = 4

Compare:

- Number of clusters
- Cluster sizes
- Cluster centroids

Record your observations.

Observation Table:

PARAMETER	OBSERVATION
Dataset Name	
Dataset File Name	
Number of Attributes	
Number of Instances	
Algorithm Used	
Number of Clusters	
Number of Iterations	
Cluster Sizes	
Cluster Centroids	
Interpretation of Clusters	

Result: The K-Means clustering algorithm was successfully applied using WEKA. Three clusters were generated from the student dataset, and their centroids and cluster distributions were analyzed. The concepts of clustering, centroids, and unsupervised learning were studied successfully.

Viva Questions:

1. What is clustering?
2. What is unsupervised learning?
3. What is K-Means?
4. What is a centroid?
5. What does K represent in K-Means?
6. How are records assigned to clusters?

7. What is the difference between clustering and classification?
8. Why are class labels not required in clustering?
9. What are the advantages of K-Means?
10. What is a cluster centroid?
11. Why may clustering results differ in different runs?
12. What is cluster interpretation?
13. What is the significance of Euclidean distance?
14. Give two applications of clustering.
15. Explain the role of K in K-Means.

Assignment Questions:

1. Explain the K-Means clustering algorithm with a suitable example.
2. Differentiate between classification and clustering.
3. What is a centroid? Explain its role in clustering.
4. Apply K-Means clustering on a dataset and interpret the generated clusters.
5. Explain the advantages and limitations of K-Means.
6. Discuss applications of clustering in recommendation systems.
7. Compare supervised and unsupervised learning.

References:

1. Ian H. Witten, Eibe Frank and Mark A. Hall, *Data Mining: Practical Machine Learning Tools and Techniques*, Morgan Kaufmann.
2. Jiawei Han, Micheline Kamber and Jian Pei, *Data Mining: Concepts and Techniques*, Morgan Kaufmann.
3. WEKA Official Documentation & User Manual.

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